

Atomic structure

Landmarks in the evolution of atomic structure are:

- 1805 Dalton's atomic theory
- 1896 Thomson discovery of electron and proton
- 1909 Rutherford's nuclear atom
- 1913 Bohr's atomic model
- 1932 Chadwick's discovery of the neutron

Bohr Model of the Atom:

1. **Concept of stationary orbit or Energy levels of Electrons:** Electrons travel around the nucleus in specific permitted circular orbits and no others.
2. **Absorption or Emission of Energy and concept of Spectrum:** In these specific orbits, an electron does not radiate (or lose) energy. When an electron goes to an orbital of the upper level, it absorbs energy and when going to an orbital of lower energy emits energy as light.
3. **Postulate Concerning Angular momentum:** The permissible orbits are those for which the angular momentum of the electron is an integral multiple of $\frac{h}{2\pi}$. The angular momentum is $mvr = \frac{nh}{2\pi}$ where $n = 1, 2, 3, \dots$
h- Planck's constant, m-mass of an atom, v-velocity, r-distance of an electron from the nucleus.

Limitations of the Bohr's Atom Model:

1. Though Bohr's Atom model can explain the Hydrogen atom spectrum, it can't explain the spectrum of atoms having several electrons.
2. For the transformation of electrons from one energy level to another one, only one line should be obtained in the spectrum, but after analyzing the spectrum by high energy spectrometer several fine lines are obtained. Bohr's atom model can't explain how these fine multiple lines are formed in the atomic spectra.
3. Bohr's theory fails to recognize the wave property of electrons, which de Broglie established.
4. According to Bohr's model electrons revolve in circular orbits but modern research reveals that electrons revolve three-dimensionally.
5. By precisely defining the radius of the atomic orbits, the theory violated Heisenberg's uncertainty principle according to which, it is impossible to define simultaneously the position and momentum of a moving sub-atomic particle. So, although Bohr's theory improved the atom model, it failed and measurably failed to explain the atom satisfactorily.

WAVE MECHANICAL CONCEPT OF ATOM

The Wave mechanical theory suggests that electron motion is of a complex nature best described by its wave properties and probabilities. While the classical 'mechanical theory' of matter considered matter to be made of discrete particles (atoms, electrons, protons, etc.). According to the wave theory, radiations as X-rays and light, consist of a continuous collection of waves traveling in space. The wave nature of light failed to explain the emission of electrons from metal surfaces by the action of light. Thereby, Planck and Einstein (1905) proposed that energy radiations are emitted discontinuously as little 'bursts', quanta, or photons.

This view is directly opposed to the wave theory of light and it gives particle-like properties to waves. According to it, light exhibits both a wave and a particle nature, under suitable conditions. This theory which applies to all radiations, is often referred to as the '**Wave Mechanical Theory**'.

In 1924 Louis de Broglie hypothesised the dual character—the wave and particle—may not be confined to radiations alone but should be extended to matter as well.

In other words, matter also possessed particle as well as wave character. This gave birth to the 'Wave mechanical theory of matter'.

THE WAVE NATURE OF ELECTRON

de Broglie's revolutionary suggestion that moving electrons had waves of definite wavelength. Later Davison and Germer (1927) demonstrated that the physical reality of the wave nature of electrons by showing that a beam of electrons could also be diffracted by crystals just like light or X-rays. They observed that the diffraction patterns thus obtained were just similar to those in case of X-rays. It was possible that electrons by their passage through crystals may produce secondary X-rays, which would show diffraction effects on the screen. Thomson ruled out this possibility, showing that the electron beam as it emerged from the crystals, underwent deflection in the electric field towards the positively charged plate

HEISENBERG'S UNCERTAINTY PRINCIPLE

One of the most important consequences of the dual nature of matter is the uncertainty principle developed by Werner Heisenberg in 1927. According to the uncertainty principle, it is impossible to know simultaneously both the conjugate properties accurately. For example, the position and momentum of a moving particle are interdependent and thus conjugate properties also. Both the position and the momentum of the particle at any instant cannot be determined with absolute exactness or certainty. If the momentum (or velocity) be measured very accurately, a measurement of the position of the particle correspondingly becomes less precise. On the other

hand if position is determined with accuracy or precision, the momentum becomes less accurately known or uncertain. Thus certainty of determination of one property introduces uncertainty of determination of the other. The uncertainty in measurement of position, Δx , and the uncertainty of determination of momentum, Δp (or Δmv), are related by Heisenberg's relationship as

$$\Delta x \times \Delta p \geq \frac{h}{2\pi}$$

Or, $\Delta x \times m\Delta v \geq \frac{h}{2\pi}$, where h is Planck's constant.

There exists a clear difference between the behaviour of large objects like a stone and small particles such as electrons. The uncertainty product is negligible in case of large objects.

For a moving ball of iron weighing 500 g, the uncertainty expression assumes the form

$$\Delta x \times m\Delta v \geq \frac{h}{2\pi}, \text{ or } \Delta x \times \Delta v \geq \frac{h}{2\pi}$$

$$\Delta x \times \Delta v \geq \frac{6.625 \times 10^{-27}}{2 \times 3.14 \times 500} \approx 5 \times 10^{-3} \text{ erg sec g}^{-1}$$

which is very small and thus negligible. Therefore, for large objects, the uncertainty of measurements is practically nil.

But for an electron of mass $m = 9.109 \times 10^{-28}$ g, the product of the uncertainty of measurements is quite large as $\Delta x \times \Delta v \geq \frac{h}{2\pi m}$ or, $\Delta x \times \Delta v \geq \frac{6.625 \times 10^{-27}}{2 \times 3.14 \times 9.109 \times 10^{-28}} \approx 0.3 \text{ erg sec g}^{-1}$

This value is large enough in comparison with the size of the electron and not negligible. If position is known quite accurately i.e., Δx is very small, the uncertainty regarding velocity Δv becomes immensely large and vice versa. It is therefore very clear that the uncertainty principle is only important in considering measurements of small particles comprising an atomic system.

de BROGLIE'S EQUATION

de Broglie had arrived at his hypothesis with the help of Planck's Quantum Theory and Einstein's Theory of Relativity. He derived a relationship between the magnitude of the wavelength associated with the mass 'm' of a moving body and its velocity. According to Planck, the photon energy 'E' is given by the equation $E = h\nu$ (i)

where h is Planck's constant and ν is the frequency of radiation. By applying Einstein's mass-energy relationship, the energy associated with a photon of mass 'm' is given as

$$E = mc^2 \dots\dots\dots(ii)$$

where c is the velocity of radiation.

$$\text{Comparing equations (i) and (ii)} \quad mc^2 = h\nu = h\frac{c}{\lambda} \quad (\because \nu = \frac{c}{\lambda})$$

$$\text{or } mc = \frac{h}{\lambda} \dots\dots\dots(iii)$$

$$\text{or mass} \times \text{velocity} = \frac{h}{\text{wavelength}}$$

The equation (iii) is called de Broglie's equation. It can be expressed as- The momentum of a particle in motion is inversely proportional to wavelength, Planck's constant 'h' being the constant of proportionality.

Schrödinger Equation

Bohr's atom model was unable to explain-

(i) Heisenberg uncertainty principle (ii) De Broglie's dual nature of electrons and (iii) the concept of spectrum using electron's stationary orbit theory.

Addressing these three problems, in 1926 Schrodinger proposed a wave mechanics equation. In this model, Schrodinger hypothesizes that – (i) electron has a dual nature that is wave and particulate character. (ii) the electron wave evolves three-dimensionally around the positively charged nucleus.

To explain the electron wave properly, Schrodinger proposed a three-dimensional wave mechanics equation, which is known as the Schrodinger wave equation. The equation is

$$\frac{\delta^2\psi}{\delta x^2} + \frac{\delta^2\psi}{\delta y^2} + \frac{\delta^2\psi}{\delta z^2} + \frac{8\pi^2m}{h^2} \times (E - V)\psi = 0$$

In this second-order differential equation, the m-mass of electrons, E-total energy of electrons, V-potential energy of electrons in a three-dimensional (x, y, z) space, h-plank's constant, ψ (psi) – the amplitude of electron waves in the three-dimensional spaces around the nucleus.

Quantum Numbers

The number, that expresses the size, shapes, and direction of the orbital from the nucleus and the spins of the electron of their own axes, are called quantum numbers.

There are four types of quantum numbers:

1. Principal quantum number (n)
2. Subsidiary quantum number (l)
3. Magnetic quantum number (m)
4. Spin quantum number (s)

Principal quantum number

This quantum number denotes the principal shell to which the electron belongs. This is also referred to as major energy level. It represents the average size of the electron cloud i.e., the average distance of the electron from the nucleus. The principal quantum number ' n ' can have non-zero, positive, integral values $n = 1, 2, 3, \dots$ increasing by integral numbers to infinity. In a polyelectron atom or ion, the electron that has a higher principal quantum number is at a higher energy level. An electron with $n = 1$ has the lowest energy and is bound most firmly to the nucleus. The letters K, L, M, N, O, P and Q are also used to designate the energy levels or shells of electrons with a n value of 1, 2, 3, 4, 5, 6, and 7 respectively. An atom has a limited number of electrons that can have the same principal quantum number and is given by $2n^2$, where n is the principal quantum number concerned. Thus,

Principal quantum number ($n =$)	1	2	3	4
Letter designation	K	L	M	N
Maximum number of electrons ($2n^2 =$)	2	8	18	32

Subsidiary/Azimuthal Quantum number (l)

The quantum number l defines the shape of the orbital occupied by the electron and the angular momentum of the electron. It is for this reason that ' l ' is sometimes referred to as an orbital or angular quantum number. For any given value of the principal quantum number n , the azimuthal quantum number l may have all integral values from 0 to ' $n - 1$ ', each of which refers to an Energy sublevel or Sub-shell. The total number of such possible sublevels in each principal level is numerically equal to the principal quantum number of the level under consideration. These sublevels are symbolized by letters s, p, d, f, etc. For example, for the principal quantum number $n = 1$, the only possible value for l is 0 i.e., there is only one possible subshell i.e. s-subshell ($n = 1, l = 0$). For $n = 2$, there are two possible values of l , $l = 0$ and

$l = 2 - 1 = 1$. This means that there are two subshells in the second energy shell with $n = 2$. These subshells are designated as 2s and 2p. Similarly, when $n = 3$, l can have three values i.e. 0, 1 and 2. Thus there are three subshells in third energy shell with designations 3s, 3p and 3d respectively. For $n = 4$, there are four possible values of azimuthal quantum number l ($= 0, 1, 2$, and 3) each representing a different sublevel. In other words, the fourth energy level consists of four subshells which are designated as 4s, 4p, 4d and 4f. Thus for different values of principal quantum numbers we have

$n = 1$ $l = 0$ (1s)	$n = 2$ $l = 0$ (2s) $l = 1$ (2p)	$n = 3$ $l = 0$ (3s) $l = 1$ (3p) $l = 2$ (3d)	$n = 4$ $l = 0$ (4s) $l = 1$ (4p) $l = 2$ (4d) $l = 3$ (4f)	$n = 5$ $l = 0$ (5s) $l = 1$ (5p) $l = 2$ (5d) $l = 3$ (5f) $l = 4$ (5g)
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For a given value of the principal quantum number the order of increasing energy for different subshells

are $s < p < d < f$ (except for the H atom).

Magnetic Quantum Number 'm'

This quantum number has been proposed to account for the splitting up of spectral lines (Zeeman Effect). An application of a strong magnetic field to an atom reveals that electrons with the same values of principal quantum number ' n ' and of azimuthal quantum number ' l ', may still differ in their behavior. They must, therefore, be differentiated by introducing a new quantum number, the magnetic quantum number m . This is also called the Orientation Quantum Number because it gives the orientation or distribution of the electron cloud. For each value of the azimuthal quantum number ' l ', the magnetic quantum number m , may assume all the integral values between $+l$ to $-l$ through zero. Therefore, for each value of l , there will be $(2l + 1)$ values of m . Thus, when $l = 0$, $m = 0$, and no other value. This means that for each value of the principal quantum number ' n ', there is only one orientation for $l = 0$ (s orbital) or there is only one s orbital. For s orbital, there being only one orientation, it must be spherically symmetrical about the nucleus. There is only one spherically symmetrical orbital for each value of n whose radius depends upon the value of n .

Spin Quantum Number 's'

This quantum number has been introduced to account for the spin of electrons about their axis. Since an electron can spin clockwise or anticlockwise (in two opposite directions), two possible values of s are equal and opposite. As quantum numbers can differ only by unity from each other, there are two values given to s ; $1/2 +$ and $1/2 -$ depending upon whether the electron spins in one direction or the other. These spins are also designated by arrows pointing upwards and downward $\downarrow \uparrow$. Two electrons with the same sign of the spin quantum numbers are said to have parallel spins while those having opposite signs of the spin quantum numbers are said to have opposite spin or antiparallel spin or paired-up spin.

Problem 1.

List all possible values of l and m for $n = 2$.

SOLUTION. Here, the principal quantum number $n = 2$. The azimuthal quantum number can have only two values. These are 0 and 1

When $l = 0$ $m = 0$
and $l = 1$ $m = +1, 0, -1$

Problem 2. What designations are given to the orbitals having

- (a) $n = 2$ $l = 1$
- (b) $n = 1$ $l = 0$
- (c) $n = 3$ $l = 2$
- (d) $n = 4$ $l = 3$

SOLUTION

- (a) when $n = 2$ and $l = 1$ the orbital is $2p$
- (b) when $n = 1$ and $l = 0$ the orbital is $1s$
- (c) when $n = 3$ and $l = 2$ the orbital is $3d$
- (d) when $n = 4$ and $l = 3$ the orbital is $4f$

Pauli Exclusion Principle

No two electrons in an atom can have the same set of four identical quantum numbers.

Even if two electrons have the same values for n , l and m , they must have different values of s .

Consider the second shell ($n = 2$), there being four orbitals, one s orbital ($l = 0$) and three p orbitals ($l = 1$), the possible number of electrons having different set of quantum numbers can be as follows :

n	l	m	s
2	0	0	$+\frac{1}{2}$
2	0	0	$-\frac{1}{2}$

Two electrons
accommodated in
2S orbital ($l = 0$)

Aufbau Principle

Electrons go to different orbitals according to their energy. Electrons first go to the orbital of lower energy and then systematically to the orbital of higher energy. It is known as Aufbau's law.

The orbital, which has a lower value of $(n+l)$ has a lower energy and a higher value of $(n+l)$ has higher energy.

Example

Between 4s and 3d orbital which is the lowest energy orbital?

For 4s orbital, $n=4$ and the value of $l=0$ (s).

The total value of $(n+l)$ is 4.

For 3d orbital, $n=3$ and the value of $l=2$ (d). The total value of $(n+l)$ is 5.

So, electrons will go to the 4s orbital first.

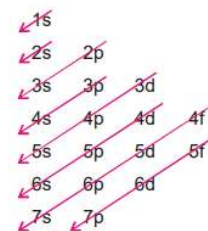


Fig. Aufbau order of orbitals for feeding in electrons.

Hund's Rule

When several orbitals of equal energy (degenerate orbitals) are available, electrons prefer to occupy separate orbitals rather than getting paired in the same orbital. Such electrons tend to have the same spins. Explanation: Two electrons with parallel spins tend to be as far apart as possible to minimize the electrostatic repulsion. Therefore, the electrons prefer to occupy the orbital singly as far as possible. When all the orbital gets singly occupied, then the incoming electron will pair up with the other electron. Orbitals having the maximum number of unpaired electrons possess the maximum stability. Half-filled and full-filled orbitals have extra stability i.e. $s^1, s^2, p^3, p^6, d^5, d^{10}, f^7, f^{14}$ are stable configuration.

$N(7)=1s^2 2s^2 2p_x^1 p_y^1 p_z^1$ (According to Hund's rule)

But, $N(7)=1s^2 2s^2 2p_x^2 p_y^1 p_z^0$ ----- not possible

Electronic Configuration

For complicated atoms which may contain many electrons and have many energy levels or orbitals, the 'building up' process for the electrons is governed by the following rules:

Rule 1. Each electron shell can hold a maximum of $2n^2$ electrons where n is the shell number.

Rule 2. These electrons are accommodated in s , p , d and f orbitals, the maximum number of electrons in each type of orbitals being determined by its electron-holding capacity (for $s = 2$, $p = 6$, $d = 10$ and $f = 14$).

Rule 3. Aufbau principle

Rule 4. Pauli's exclusion principle

Rule 5. Hund's rule

TABLE 2.2. ANOMALOUS ELECTRON CONFIGURATION OF SOME ELEMENTS			
Element	At. No.	Expected Configuration	Actual Configuration
Cr	24	$4d^4 4s^2$	$3d^5 4s^1$
Cu	29	$3d^9 4s^2$	$3d^{10} 4s^1$
Mo	42	$4d^4 5s^2$	$4d^5 5s^1$
Pd	46	$4d^8 5s^2$	$4d^{10} 5s^0$
Ag	47	$4d^9 5s^2$	$4d^{10} 5s^1$
Pt	78	$5d^8 6s^2$	$5d^9 6s^1$
Au	79	$5d^9 6s^2$	$5d^{10} 6s^1$

TABLE 2.3. GROUND STATE ELECTRON CONFIGURATION OF ELEMENTS					
Z	Element	Electron configuration	Z	Element	Electron configuration
1	H	$1s^1$	21	Sc	$[\text{Ar}]3d^14s^2$
2	He	$1s^2$	22	Ti	$[\text{Ar}]3d^24s^2$
3	Li	$[\text{He}]2s^1$	23	V	$[\text{Ar}]3d^34s^2$
4	Be	$[\text{He}]2s^2$	24	Cr	$[\text{Ar}]3d^54s^1$
5	B	$[\text{He}]2s^22p^1$	25	Mn	$[\text{Ar}]3d^54s^2$
6	C	$[\text{He}]2s^22p^2$	26	Fe	$[\text{Ar}]3d^64s^2$
7	N	$[\text{He}]2s^22p^3$	27	Co	$[\text{Ar}]3d^74s^2$
8	O	$[\text{He}]2s^22p^4$	28	Ni	$[\text{Ar}]3d^84s^2$
9	F	$[\text{He}]2s^22p^5$	29	Cu	$[\text{Ar}]3d^{10}4s^1$
10	Ne	$[\text{He}]2s^22p^6$	30	Zn	$[\text{Ar}]3d^{10}4s^2$
11	Na	$[\text{Ne}]3s^1$	31	Ga	$[\text{Ar}]3d^{10}4s^24p^1$
12	Mg	$[\text{Ne}]3s^2$	32	Ge	$[\text{Ar}]3d^{10}4s^24p^2$
13	Al	$[\text{Ne}]3s^23p^1$	33	As	$[\text{Ar}]3d^{10}4s^24p^3$
14	Si	$[\text{Ne}]3s^23p^2$	34	Se	$[\text{Ar}]3d^{10}4s^24p^4$
15	P	$[\text{Ne}]3s^23p^3$	35	Br	$[\text{Ar}]3d^{10}4s^24p^5$
16	S	$[\text{Ne}]3s^23p^4$	36	Kr	$[\text{Ar}]3d^{10}4s^24p^6$
17	Cl	$[\text{Ne}]3s^23p^5$	37	Rb	$[\text{Kr}]5s^1$
18	Ar	$[\text{Ne}]3s^23p^6$	38	Sr	$[\text{Kr}]5s^2$
19	K	$[\text{Ar}]4s^1$	39	Y	$[\text{Kr}]4d^15s^2$
20	Ca	$[\text{Ar}]4s^2$	40	Zr	$[\text{Kr}]4d^25s^2$

Z	Element	Electron configuration	Z	Element	Electron configuration
41	Nb	$[\text{Kr}]4d^45s^1$	73	Ta	$[\text{Xe}]4f^{14}5d^36s^2$
42	Mo	$[\text{Kr}]4d^55s^1$	74	W	$[\text{Xe}]4f^{14}5d^46s^2$
43	Tc	$[\text{Kr}]4d^55s^2$	75	Re	$[\text{Xe}]4f^{14}5d^56s^2$
44	Ru	$[\text{Kr}]4d^75s^1$	76	Os	$[\text{Xe}]4f^{14}5d^66s^2$
45	Rh	$[\text{Kr}]4d^85s^1$	77	Ir	$[\text{Xe}]4f^{14}5d^76s^2$
46	Pd	$[\text{Kr}]4d^{10}$	78	Pt	$[\text{Xe}]4f^{14}5d^96s^1$
47	Ag	$[\text{Kr}]4d^{10}5s^1$	79	Au	$[\text{Xe}]4f^{14}5d^{10}6s^1$
48	Cd	$[\text{Kr}]4d^{10}5s^2$	80	Hg	$[\text{Xe}]4f^{14}5d^{10}6s^2$
49	In	$[\text{Kr}]4d^{10}5s^25p^1$	81	Tl	$[\text{Xe}]4f^{14}5d^{10}6s^26p^1$
50	Sn	$[\text{Kr}]4d^{10}5s^25p^2$	82	Pb	$[\text{Xe}]4f^{14}5d^{10}6s^26p^2$
51	Sb	$[\text{Kr}]4d^{10}5s^25p^3$	83	Bi	$[\text{Xe}]4f^{14}5d^{10}6s^26p^3$
52	Te	$[\text{Kr}]4d^{10}5s^25p^4$	84	Po	$[\text{Xe}]4f^{14}5d^{10}6s^26p^4$
53	I	$[\text{Kr}]4d^{10}5s^25p^5$	85	At	$[\text{Xe}]4f^{14}5d^{10}6s^26p^5$
54	Xe	$[\text{Kr}]4d^{10}5s^25p^6$	86	Rn	$[\text{Xe}]4f^{14}5d^{10}6s^26p^6$
55	Cs	$[\text{Xe}]6s^1$	87	Fr	$[\text{Rn}]7s^1$
56	Ba	$[\text{Xe}]6s^2$	88	Ra	$[\text{Rn}]7s^2$
57	La	$[\text{Xe}]5d^16s^2$	89	Ac	$[\text{Rn}]6d^17s^2$
58	Ce	$[\text{Xe}]4f^15d^16s^2$	90	Th	$[\text{Rn}]6d^27s^2$
59	Pr	$[\text{Xe}]4f^36s^2$	91	Pa	$[\text{Rn}]5f^26d^17s^2$
60	Nd	$[\text{Xe}]4f^46s^2$	92	U	$[\text{Rn}]5f^36d^17s^2$
61	Pm	$[\text{Xe}]4f^56s^2$	93	Np	$[\text{Rn}]5f^46d^17s^2$
62	Sm	$[\text{Xe}]4f^66s^2$	94	Pu	$[\text{Rn}]5f^67s^2$
63	Eu	$[\text{Xe}]4f^76s^2$	95	Am	$[\text{Rn}]5f^77s^2$
64	Gd	$[\text{Xe}]4f^75d^16s^2$	96	Cm	$[\text{Rn}]5f^76d^17s^2$
65	Tb	$[\text{Xe}]4f^96s^2$	97	Bk	$[\text{Rn}]5f^97s^2$
66	Dy	$[\text{Xe}]4f^{10}6s^2$	98	Cf	$[\text{Rn}]5f^{10}7s^2$
67	Ho	$[\text{Xe}]4f^{11}6s^2$	99	Es	$[\text{Rn}]5f^{11}7s^2$
68	Er	$[\text{Xe}]4f^{12}6s^2$	100	Fm	$[\text{Rn}]5f^{12}7s^2$
69	Tm	$[\text{Xe}]4f^{13}6s^2$	101	Md	$[\text{Rn}]5f^{13}7s^2$
70	Yb	$[\text{Xe}]4f^{14}6s^2$	102	No	$[\text{Rn}]5f^{14}7s^2$
71	Lu	$[\text{Xe}]4f^{14}5d^16s^2$	103	Lr	$[\text{Rn}]5f^{14}6d^17s^2$
72	Hf	$[\text{Xe}]4f^{14}5d^26s^2$	104	Rf	$[\text{Rn}]5f^{14}6d^27s^2$

Note : The symbol in brackets indicates the electron core of the Noble gas.